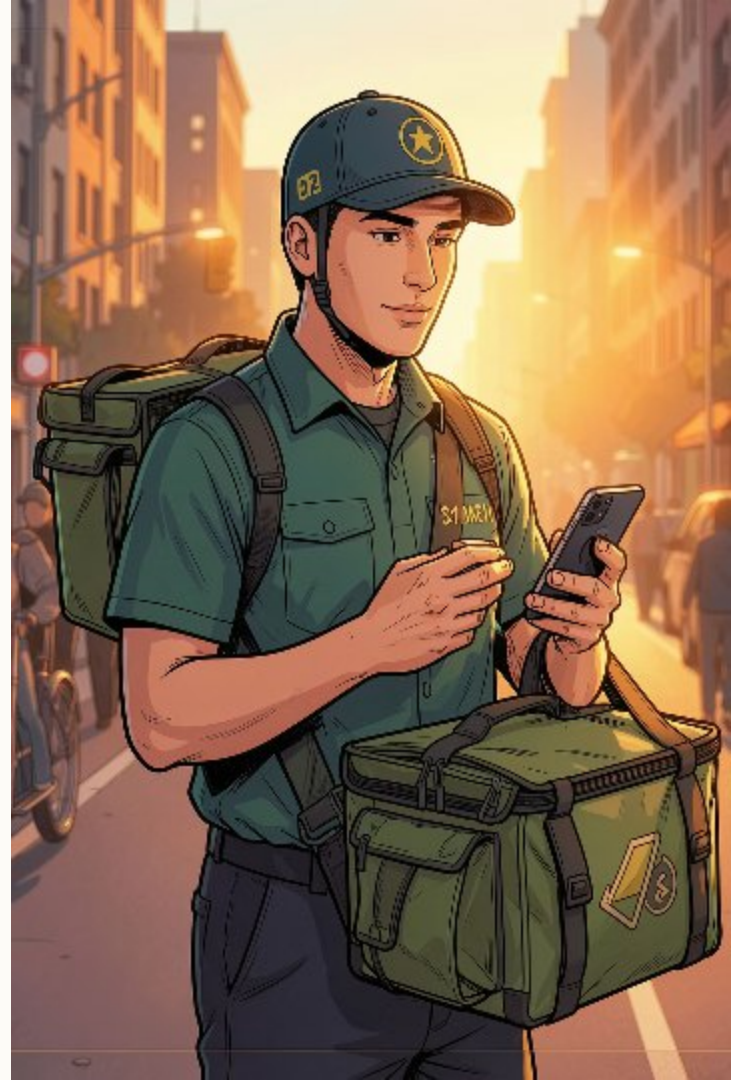


Food Express: Delivery & Sales Analytics

End-to-end data analytics portfolio project –Python, SQL, Power BI. A concise narrative from business problem to dashboard-ready insights, prepared by Kumud.

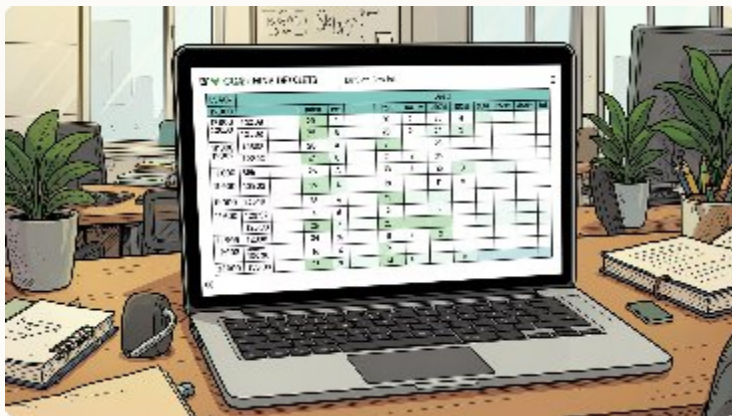




FoodExpress lacks a unified performance view—data is fragmented across systems. The objective: consolidate and analyze order, delivery, cancellation, payment, and rating data to increase revenue and delivery efficiency.

- Which cities & restaurants drive the most revenue?
- What factors affect delivery time and ratings?
- Why are orders cancelled and how to reduce rate?
- Customer preferences cuisines payment methods top customers

Dataset Overview



Quick facts

- 520 orders, 24 columns
- 7 cities: Mumbai, Delhi, Bangalore, Hyderabad, Pune, Chennai, Kolkata
- 24 restaurants, 10 cuisine types
- ~7 months of order history including order metadata, customer profiles, delivery logistics, payment details, and ratings

Data readiness: minimal structural issues; primary work focused on null handling, timestamp parsing, and categorical standardization.

Methodology

Three-phase approach aligned to stakeholder questions and deployable outputs.

01

1—Python EDA & Cleaning

Use Pandas for null analysis, type conversion, feature engineering (order_hour, prep_time buckets), and Seaborn/Matplotlib for distribution & correlation checks.

03

3—Power BI Dashboard

Two interactive pages: Operational KPIs (delivery SLA, cancellations, time-series) and Revenue & Customers (geography, top restaurants, LTV candidates). Live connection to Postgres.

02

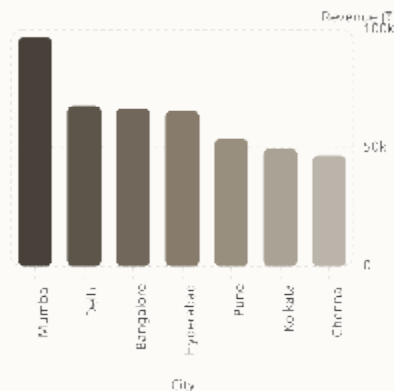
2—SQL Analysis (Postgres)

Author reproducible queries for revenue by city/restaurant, cancellation drivers (CASE WHEN), delivery time correlations, and cohort-level customer value.



Key Finding —Revenue by City

Headline: Mumbai leads with ₹97,000 in total revenue; Delhi, Bangalore, and Hyderabad follow closely (≈₹66-68K). Chennai records the lowest (≈₹47K).

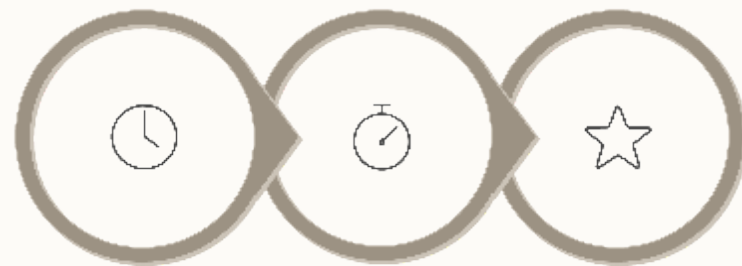


- Concentration: top 2 cities account for the majority of platform revenue.
- Opportunity: replicate Mumbai tactics (high-performing restaurants, promotional cadence) in mid-tier cities



Key Finding –Delivery Time & Ratings

Average delivery time cluster: ~31-33 minutes. Dinner slot has the highest mean (~33 mins), likely due to order volume and kitchen load-correlates with slightly lower average ratings



Peak Order
Windows

Longer Delivery
Times

Slight Rating Drop

Recommendation: test dynamic batching and time-windowed incentives for dinner hours; monitor rating impact over a 4-week pilot.

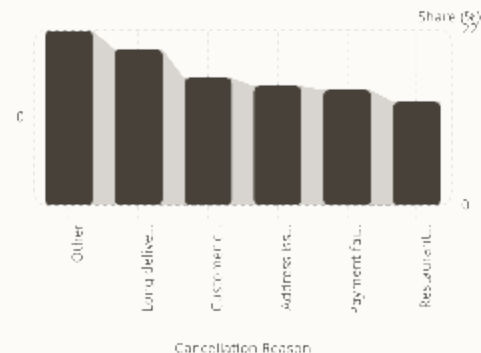


Key Finding—Cancellations

Overall cancellation rate: 10.77% Top cancellation reason: long delivery time (19.6%). Other causes (customer change, address issues, payment failures, stock-out-s) are distributed evenly—no single fix will eliminate cancellations.

59.36%

Total conversion rate



Short-term fixes

Auto-reassign slow orders,
estimated-time accuracy
improvements, proactive
customer alerts

Medium term

Optimize dispatch algorithm,
partner-level SLA scorecards,
menu-level prep-time
estimation.

Long-term

Demand forecasting for dinner
windows, strategic restaurant
recruitment to increase
capacity.

Customer Preferences & Top Customers

Preferences:

- Top cuisines by order volume: North Indian > Fast Food > Chinese.
- Payment methods: UPI dominates (40.2%), COD second (20.8%) indicating strong digital adoption with a sizable cash tail.

Top customers:



Karan Sharma

Top spender - ₹3,097.64. High frequency, high recency. candidate for VIP loyalty program



Top 10 cohort

Top 10 customers contributed ₹26,008 (concentrated revenue; consider targeted retention offers).

